

## Activity:

### Identity Theft and unauthorized Access

1. Explain identity Theft in cloud computing.

Identity theft in cloud computing occurs when an attacker steals a user's login credentials (such as username, password, or authentication token) and uses them to access cloud services without permission. This can lead to data theft, financial loss, privacy violations and misuse of cloud resource.

Eg:- A hacker steals an employee's cloud account password and gains access to confidential company files stored on Google Drive or AWS.

2. Analyze How Weak passwords and phishing Attacks lead to unauthorized Access.

Weak passwords:-

- Easy-to-guess passwords (eg., 123456) can be cracked using brute-force or dictionary attacks.
- Reusing the same password across multiple accounts increases security risks.
- Attackers who obtain the password can log into cloud accounts and steal sensitive information.

## phishing Attacks:-

- Attackers send fake emails or messages pretending to be Trusted organization.
- Users are tricked into entering their login credentials on fake websites.
- The stolen credentials are then used to access cloud services without authorization.

## Impact:-

- Data Theft
- Financial fraud
- Loss of privacy
- unauthorized modification or deletion of cloud data.

## 3. Recommend Authentication Mechanisms

### Multi - Factor Authentication (MFA)

- Requires two or more verification methods.
- Eg:- Password + OTP sent to a mobile phone or finger print scan.
- Even if the password is stolen, attackers cannot access the account without the second factor.

### Strong password policy:-

- Use at least 12-16 characters
- Include uppercase letters, lowercase, numbers and special symbols.
- Avoid personal information (name, birthday, phone number).